Computing p-value from the 2D Kolmogorov-Smirnov Statistic [1]

The required input to calculate the chance probability or p-value of a 2-dimensional KS-Test statistic in the comparison of two empirical data sets is:

- d_1: The supremum of the comparison using data set 1 as origin of quadrants.
- d_2: The supremum of the comparison using data set 2 as origin of quadrants.
- n_1: The number of data points in data set 1.
- n_2: The number of data points in data set 2.
- r_1: The Pearson Correlation Coefficient of data set 1.
- r_2: The Pearson Correlation Coefficient of data set 2.

We then find the p-value as follows:
Compute D, the average supremum.

$$D = \frac{d_1 + d_2}{2}$$

Compute N, the effective number of data points in the comparison.

$$N = \frac{n_1 n_2}{n_1 + n_2}$$

Combine the Pearson Coefficients into a single number r

$$r = \sqrt{1 - \frac{r_1^2 + r_2^2}{2}}$$

Then assemble λ, the input to the p-value summation.

$$\lambda = \frac{\sqrt{N} D}{1 + r\left(\frac{1}{4} - \frac{3}{4}\frac{1}{\sqrt{N}}\right)}$$

Finally, the p-value is given by

$$p = \Pr(D \geq D_{obs}) = 2\sum_{k=1}^{\infty}(-1)^{k-1} e^{-2k^2\lambda^2}$$

Carrying the sum to $k = 10$ (note that the summation starts with $k = 1$) should be plenty.

[1] W.H. Press and S.A. Teukolsky, *Computers in Physics* **2** 74 (1988).